

WILP, BITS Pilani



MATHEMATICAL FOUNDATIONS FOR MACHINE LEARNING · COMPANION READER

Constrained Optimization

Lagrangian, slack variables, the KKT conditions, and the primal–dual pair

Live session, 26 July 2026 · Read alongside the lecture · Every board example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

How to use this companion. The lecture gave you the board work; this reader gives you the *why*. Every idea from the session is expanded into plain English. Every board example is worked out in full, with no skipped steps.

Colour-coded boxes guide you. The **blue** “intuition” box states the core idea in one breath. The **amber** “everyday picture” box ties it to real life, before any math. The **green** “worked example” box solves a problem with real numbers. The **red** “watch out” box flags the common mistake. The **teal** “wait — really?” box holds a true fact that sounds wrong. The **purple** “key takeaway” box is the one line to remember. Read it in order.

1 Solve one right now, before any theory

Put a wall in front of gradient descent and the answer moves onto the wall. That single fact drives this whole session.

Here is a problem you can do in your head. Find the smallest value of $f(x) = x^2$, but you are only allowed to pick $x \geq 2$.

Try it. The bowl x^2 has its bottom at $x = 0$. That point is banned. So you slide as close to it as the rule lets you. That is $x = 2$, giving $f = 4$.

Now notice three things about what you just did.

1. The free answer, $x = 0$, was thrown away. The rule changed the answer.
2. The new answer sat *exactly on the edge* of the allowed set.
3. At $x = 2$ the slope of f is 4, not 0. The usual “set the derivative to zero” test fails here.

Point 3 is the reason this session exists. Every method you have learnt so far hunts for a zero gradient. With a rule in the way, the winning point can have a non-zero gradient. We need a new test. That test is called KKT, and by the end of this reader you will be able to run it.

★ Everyday picture

Think of a hike in a national park. You want to reach the lowest point in the valley. A sign says: do not cross this fence. The true bottom lies past the fence. So the best you can do is walk right up to the fence and stop. We will use this one story — *the park, the fence, and a guard who fines you* — for the whole reader.

✓ Key takeaway

A constrained problem is an unconstrained problem with a wall added. The answer usually ends up pressed against the wall.

A fence moves the answer onto its own edge

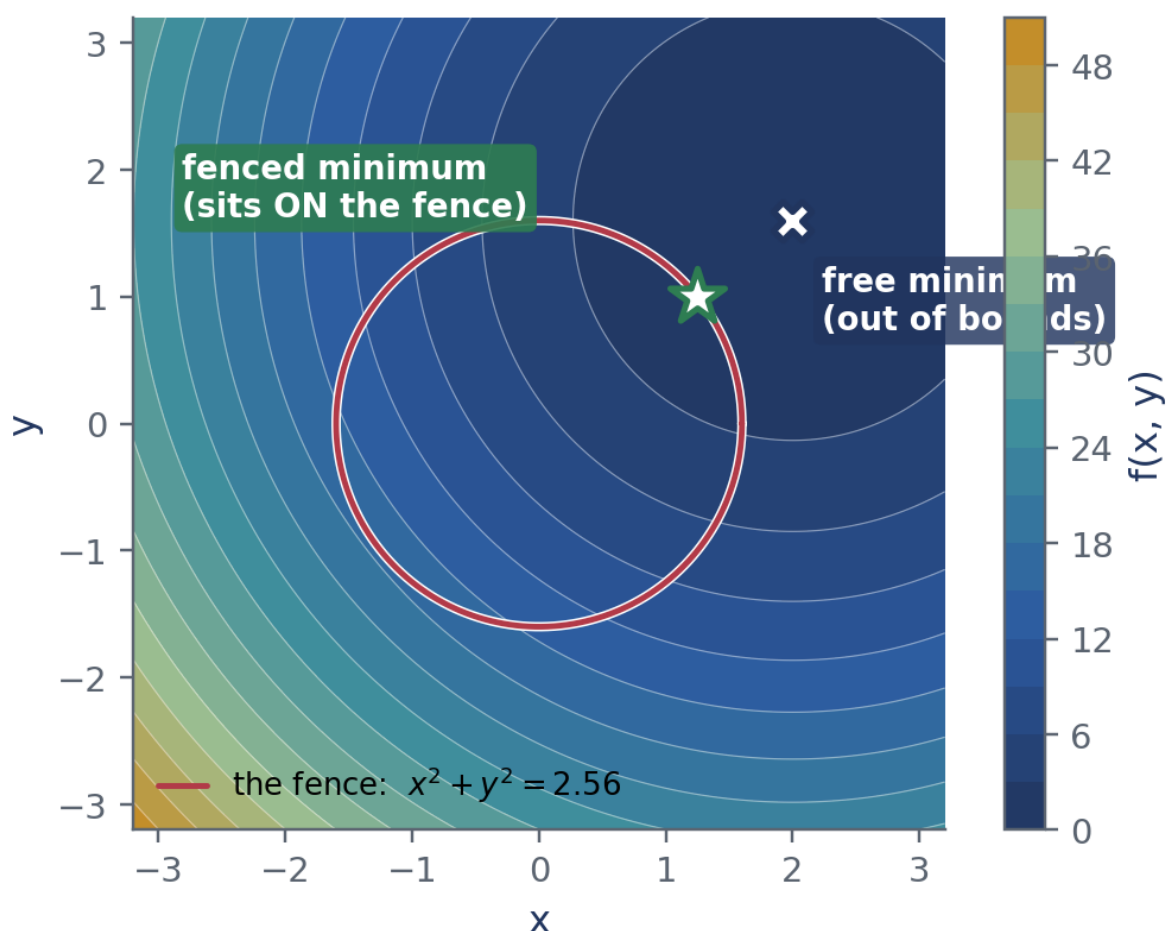


Figure 1: How to read it: the colours are the value of f , dark is low. The white-and-red circle is the fence. The X marks the free minimum, which is out of bounds. The star is the real answer, and it sits on the fence, not inside it.

2 The standard form: one shape for every problem

Every constrained problem is rewritten as “minimise f , subject to $e = 0$ and $g \leq 0$ ”. Two lines of algebra get any problem into that shape.

A **constrained optimization problem** has an *objective* (the thing you want smallest) and *constraints* (the rules you must obey). We always write it in one fixed layout, called the **standard form**:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & e(\mathbf{x}) = 0 \\ & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \end{aligned} \tag{1}$$

Read the symbols one at a time.

- $\mathbf{x}$ (bold $\mathbf{x}$) is the whole list of unknowns, such as (x, y) or $(\beta_1, \dots, \beta_p)$.
- $f(\mathbf{x})$ is the **objective function**: the score we want as small as possible. We will call it *the score*.
- $e(\mathbf{x}) = 0$ is an **equality constraint**. It is a rule you must sit exactly on. We will call it *the path*. Most exam problems have just one.
- $g_i(\mathbf{x}) \leq 0$ are the **inequality constraints**. Each one is a wall you must stay behind. We will call each one *a fence*. There can be many, so we number them $i = 1, \dots, m$.

2.1 Turning any rule into standard form

Your problem will rarely arrive in this shape. Two moves fix that.

Move 1: flip a $\geq$ into a $\leq$. Multiply both sides by -1 and the arrow turns around:

$$h(\mathbf{x}) \geq 0 \quad \Longleftrightarrow \quad -h(\mathbf{x}) \leq 0.$$

So $x \geq 0$ becomes $-x \leq 0$. Same rule, standard shape.

Move 2: move everything to the left. A rule like $x + y \leq 1$ becomes $x + y - 1 \leq 0$. A rule like $x + y = 1$ becomes $x + y - 1 = 0$.

Worked example: Rewrite one problem in standard form

Take the problem: minimise $x^2 + y^2$ subject to $x + y = 1$, $x \geq 0$ and $y \geq 0$.

Step 1. The score is already fine: $f(x, y) = x^2 + y^2$.

Step 2. The path: $x + y = 1$ becomes $e(x, y) = x + y - 1 = 0$.

Step 3. First fence: $x \geq 0$ becomes $g_1(x, y) = -x \leq 0$.

Step 4. Second fence: $y \geq 0$ becomes $g_2(x, y) = -y \leq 0$.

So we have one path and two fences, $m = 2$. That is the whole conversion.

✗ Watch out

When you flip $x \geq 0$ into $-x \leq 0$, the *constraint function* is $g_1 = -x$, not x . Later you will need $\nabla g_1 = (-1, 0)$, with the minus sign. Getting this sign wrong flips the sign of the fine rate and breaks the whole answer.

2.2 Why we never write a strict $<$

Look at the list again. You see $=$, $\leq$ and $\geq$. You never see $<$ or $>$. There is a good reason.

Try to minimise $f(x) = x$ subject to $x > 2$. Is $x = 2.1$ the answer? No, because 2.01 is allowed and smaller. Is 2.01 the answer? No, because 2.001 is allowed and smaller. You can keep going forever.

The values get closer and closer to 2, but 2 itself is banned. So there is no smallest allowed value. The problem has no answer at all.

With $x \geq 2$ the endpoint is allowed, so the answer is $x = 2$, and $f = 2$. Including the boundary is what makes the answer exist.

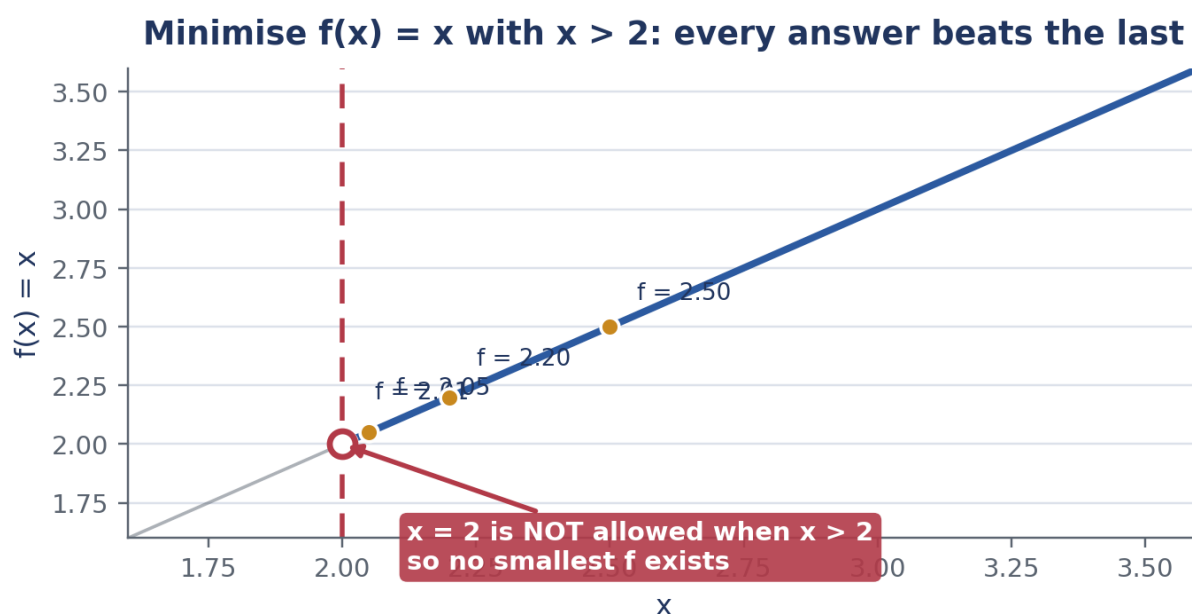


Figure 2: How to read it: the dashed red line is the banned edge $x = 2$. The amber dots are allowed points, each one beating the last. The hollow circle marks the point that would win, but is not allowed.

*** Wait — really?**

This is not a small technical point. It is the reason the answer lands on the boundary so often. Most constrained answers sit *exactly* on a fence, so a form that excludes the fence would throw the answer away.

✓ Key takeaway

Standard form is: minimise f , subject to $e = 0$ and every $g_i \leq 0$. Flip a $\geq$ by multiplying by -1 . Never use a strict $<$.

3 You have been doing this since ridge regression

Ridge and lasso are constrained problems in disguise. The penalty term you already write is the Lagrangian of a fence on the coefficients.

Before the theory, here is where you have already met this shape.

Ridge regression is usually taught as: minimise the mean squared error plus a penalty. Its honest original form is a constrained problem:

$$\begin{aligned} \min_{\beta} \quad & \text{MSE} = \frac{1}{n} \sum_{k=1}^n (y_k - \hat{y}_k)^2 \\ \text{subject to} \quad & \sum_{j=1}^p \beta_j^2 \leq s. \end{aligned} \quad (2)$$

Here β_j is the coefficient on feature j , and s is a budget you choose. **Lasso regression** is the same, with $\sum_j |\beta_j| \leq s$ instead.

Read (2) in the park story. The score is the training error. The fence is a budget on how big the coefficients may grow.

3.1 Why we fence the coefficients at all

Without gradient descent, linear regression solves for the coefficients in one shot:

$$\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y},$$

where X is the feature matrix and $\mathbf{y}$ is the output vector. The inverse is the weak spot. An inverse is the adjoint divided by the determinant. When the determinant is near zero, the entries of the inverse blow up.

The determinant goes to zero when two feature columns carry the same information. Say x_2 is exactly twice x_1 :

$$X^\top X = A = \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}, \quad \det A = 2 \cdot 8 - 4 \cdot 4 = 0.$$

The rank is 1, not 2, so A^{-1} does not exist. Real data is rarely this perfect, but with strong **multicollinearity** (features that move together) the determinant gets tiny. Then the coefficients explode. This is called **parameter inflation**. A true relationship like

$$y = 0.2x_1 + 4x_2 - 0.6x_3$$

can come back fitted as

$$\hat{y} = 200x_1 - 180x_2 + 0.79x_3.$$

The two big numbers nearly cancel. The fit looks fine on the training data and falls apart on new data.

3.2 The trick most textbooks skip

Ridge came from a plain matrix repair, not from a penalty. Add a small number λ to each diagonal entry:

$$A + \lambda I = \begin{pmatrix} 2+2 & 4 \\ 4 & 8+2 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 10 \end{pmatrix}, \quad \det = 4 \cdot 10 - 4 \cdot 4 = 24.$$

Column two is no longer exactly twice column one. The determinant is now 24, not 0, so the inverse behaves. That gives the ridge formula

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}.$$

Now the connection. Put a fine of λ on the fence in (2):

$$\begin{aligned} \mathcal{L}(\beta, \lambda) &= \text{MSE} + \lambda \left(\sum_j \beta_j^2 - s \right) \\ &= \text{MSE} + \underbrace{\lambda \sum_j \beta_j^2}_{\text{the form you know}} - \lambda s. \end{aligned}$$

The last piece, λs , does not contain β . It cannot change which β wins, so we drop it. What is left is exactly the penalised form from your regularization notes. Minimising it gives the $+\lambda I$ above. Same λ , three different disguises.

*** Wait — really?**

The λ in “ $\text{MSE} + \lambda \sum \beta_j^2$ ”, the λ added to the diagonal of $X^\top X$, and the fine rate we are about to define are the same number. Most books show one of the three and never say they are linked.

✓ Key takeaway

Ridge and lasso are constrained problems with a budget on the coefficients. The penalty you already write is that budget, priced as a fine.

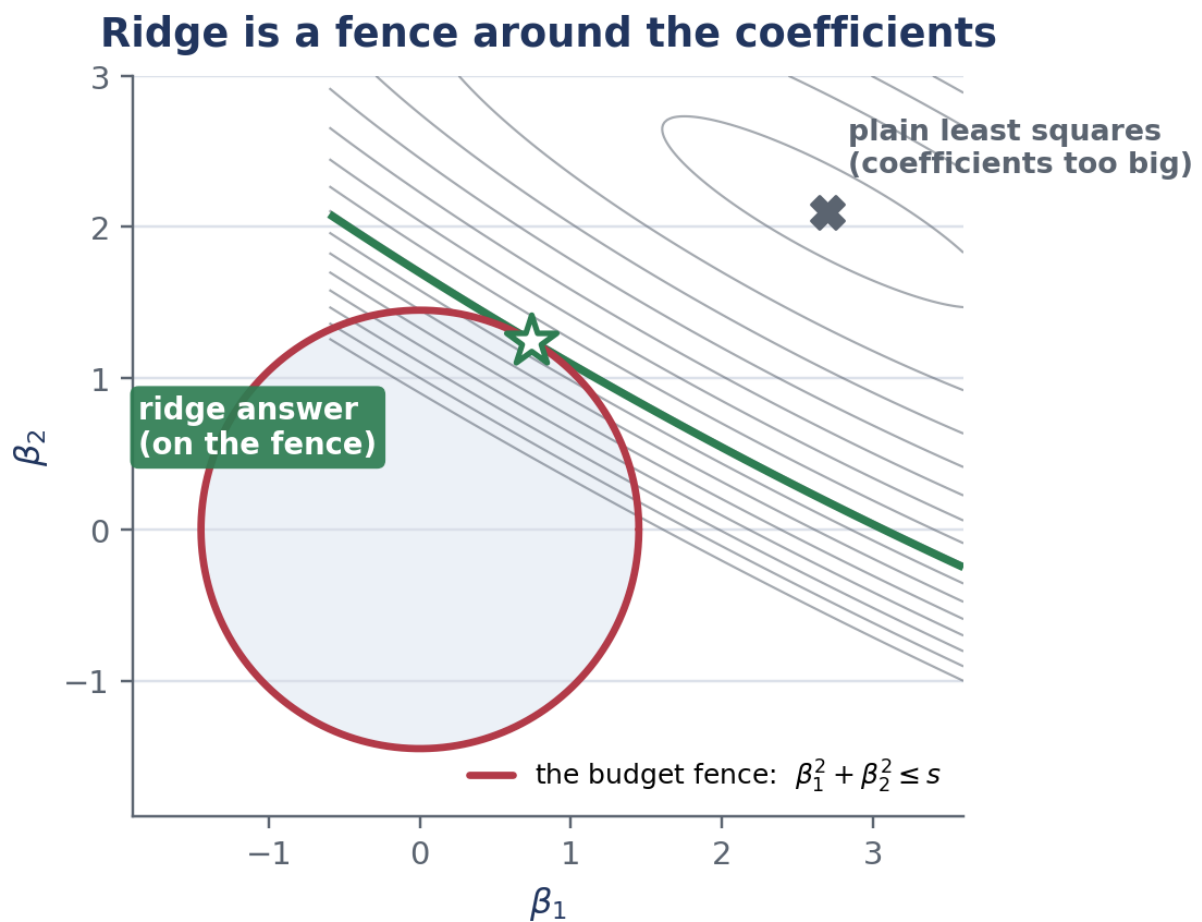


Figure 3: How to read it: grey rings are level curves of the training error over two coefficients, best at the grey X. The red circle is the budget $\beta_1^2 + \beta_2^2 \leq s$. Ridge picks the star, where the smallest reachable ring touches the budget circle.

4 The one picture: at the answer, two arrows lie on one line

Where the walk stops, the gradient of the score and the gradient of the fence point along the same line. That single geometric fact becomes the first KKT condition.

Go back to the park. You are walking downhill and you hit the fence. Why do you stop *here* and not slide further along it?

Because sliding along the fence no longer takes you downhill. At the stopping point, the downhill direction of the score points straight into the fence. There is no piece of it left that runs *along* the fence.

Say that with gradients. The gradient ∇f points uphill on the score. The gradient ∇g points out of the allowed side of the fence. “No downhill left along the fence” means these two arrows lie on one line:

$$\nabla f(\mathbf{x}^*) = -\mu \nabla g(\mathbf{x}^*), \quad \mu \geq 0. \quad (3)$$

⚠ Watch out

The two arrows are *parallel*, not equal. They almost never have the same length. The number μ is the stretch factor between them. Saying “ $\nabla f = \nabla g$ ” is the single most common slip on this topic.

🧠 The intuition

Break ∇f into two pieces: one along the fence, one across it. If the along-the-fence piece is not zero, you can still walk downhill, so you are not finished. You stop only when that piece vanishes. Then all of ∇f points across the fence, which is what (3) says.

✓ Key takeaway

At the winning point, the score’s gradient and the fence’s gradient are parallel. The stretch factor between them is the fine rate.

At the winning point the two arrows lie on one line

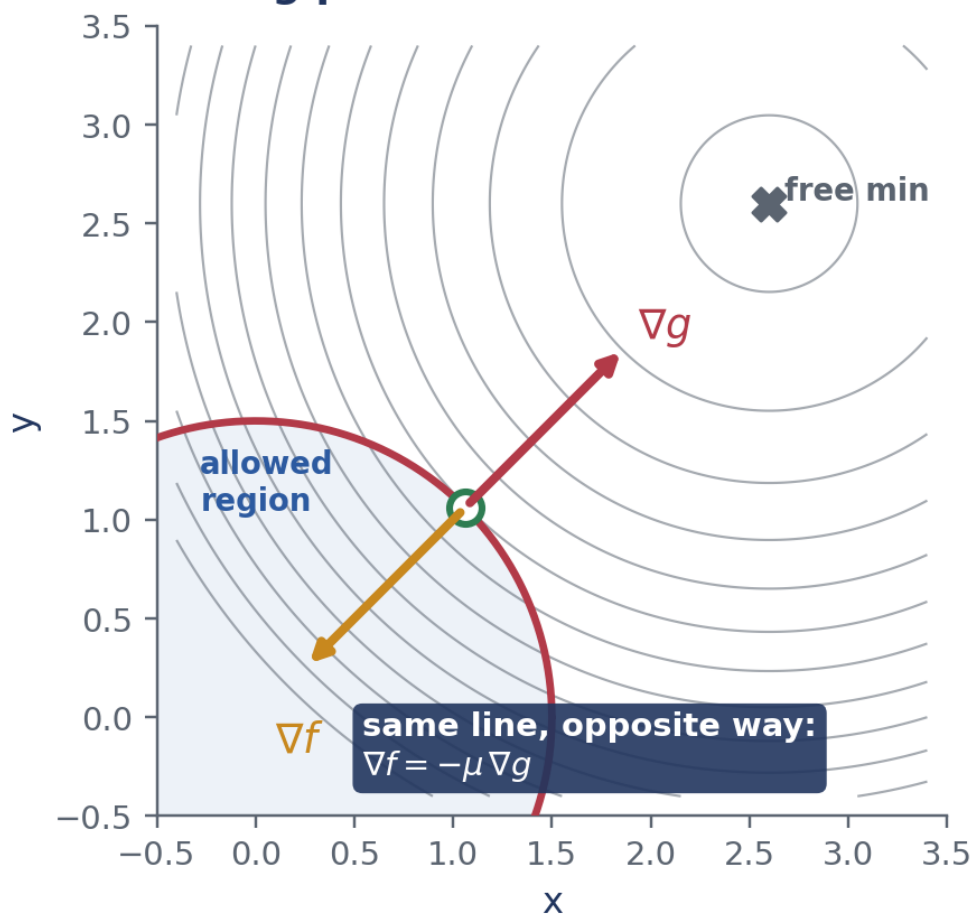


Figure 4: How to read it: grey rings are level curves of the score, the red circle is the fence, the shaded disc is the allowed region. At the white dot the amber arrow ∇f and the red arrow ∇g lie along one line, pointing opposite ways.

The ball rolls downhill until the wall stops it

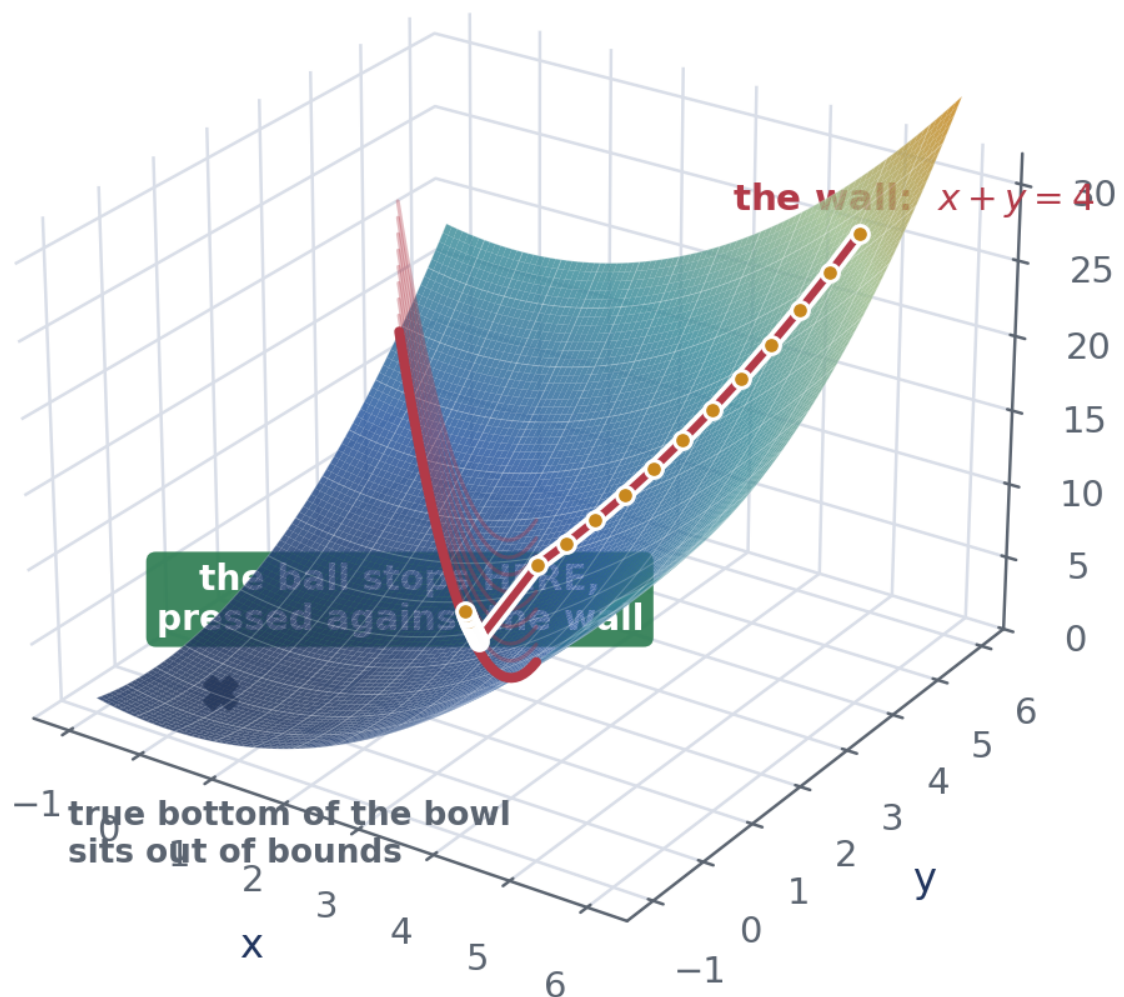


Figure 5: The same story in three dimensions. The ball rolls down the bowl and is stopped by the wall $x + y = 4$. The true bottom of the bowl is out of bounds, so the ball settles on the wall.

5 The Lagrangian: turn every fence into a fine

Add each broken rule to the score, priced by a fine rate. The rules disappear from the problem and reappear inside one unconstrained function.

We know how to optimise a function with no rules: take derivatives, set them to zero. So the plan is to *remove* the rules. The way to remove a rule is to charge for breaking it.

Build one new function, the **Lagrangian**:

$$\mathcal{L}(\mathbf{x}, \lambda, \boldsymbol{\mu}) = f(\mathbf{x}) + \lambda e(\mathbf{x}) + \sum_{i=1}^m \mu_i g_i(\mathbf{x}). \quad (4)$$

Reading it in plain words: the total bill equals the score, plus a fine for leaving the path, plus a fine for crossing each fence.

The new symbols are the **Lagrange multipliers**, also called the *fine rates*:

- λ (the Greek letter lambda) is the fine rate for the path.
- μ_i (the Greek letter mu) is the fine rate for fence i .

★ Everyday picture

A guard stands at the fence. Cross it and you pay μ rupees per metre past it. Stay behind it and you pay nothing. The guard would love you to stray, so the guard sets the fine as high as it can go. You want your bill low. That tug-of-war is the whole method.

5.1 The sign rules, and why they are not arbitrary

Two rules govern the fine rates. They look small and they decide everything.

Rule A: λ is free. It may be positive, negative or zero. The path constraint is $e(\mathbf{x}) = 0$. You can drift off it in either direction. Both directions are equally illegal, so the fine has no natural sign. We check this with real numbers in Section 10.

Rule B: every μ_i must satisfy $\mu_i \geq 0$. Here is the argument. A fine has to *raise* your bill. Suppose you cross fence i , so $g_i(\mathbf{x}) > 0$. The term added is $\mu_i g_i$. If μ_i were negative, that term would be negative. Crossing the fence would lower your bill. The optimiser would then run straight out of bounds, on purpose. Forcing $\mu_i \geq 0$ blocks that. This rule is called **dual feasibility**.

✗ Watch out

Do not carry Rule A over to a fence. For an inequality, a negative μ_i does not mean “a different sign convention”. It means the case you are testing is impossible, and you throw it away. You will use this to kill three of four cases in Section 9.

✓ Key takeaway

The Lagrangian is score + fine for the path + fines for the fences. The path fine has any sign. Every fence fine must be zero or positive.

The KKT recipe, start to finish



Figure 6: The whole method on one line. The next three sections build the middle two boxes; Section 9 runs the last two on a real problem.

6 The slack variable: how $\leq$ becomes $=$

Add a squared number to soak up the leftover room. A squared number is never negative, so the new equation says exactly what the old inequality said.

The Lagrangian in (4) treats the fence like the path. But they are not the same. The path is an equation. The fence is an inequality. Derivatives handle equations well and inequalities badly.

So we convert the fence into an equation. Take the fence $g_i(\mathbf{x}) \leq 0$ and add a new unknown t_i :

$$g_i(\mathbf{x}) + t_i^2 = 0. \quad (5)$$

The number t_i is a **slack variable**. Read it as *the room you have left before you touch fence i* .

Check that (5) really says the same thing. Rearranging, $t_i^2 = -g_i(\mathbf{x})$. A square is never negative, so $-g_i(\mathbf{x}) \geq 0$, which is $g_i(\mathbf{x}) \leq 0$. That is the original fence, back again. Nothing was lost and nothing was added.

The intuition

Why square it, instead of writing $g_i + t_i = 0$ with $t_i \geq 0$? Because then we would still be carrying an inequality, $t_i \geq 0$. The square builds “never negative” into the algebra itself. It also keeps the Lagrangian smooth, so we can differentiate it.

Now rewrite the Lagrangian with the slack terms folded in:

$$\mathcal{L}(\mathbf{x}, \lambda, \boldsymbol{\mu}, \mathbf{t}) = f(\mathbf{x}) + \lambda e(\mathbf{x}) + \sum_{i=1}^m \mu_i [g_i(\mathbf{x}) + t_i^2]. \quad (6)$$

Every rule in the problem is now an equation. There are no inequalities left, apart from the sign rule $\mu_i \geq 0$.

6.1 Count the unknowns

It pays to know how many things you are solving for.

Unknown	How many	What it is
$\mathbf{x}$	n	the variables you actually care about
λ	1	the fine rate for the path
$\mu_1, \dots, \mu_m$	m	one fine rate per fence
$t_1, \dots, t_m$	m	one room-left number per fence
Total	$n + 2m + 1$	

Books that skip slack variables report $n + m + 1$ instead. Both counts are right. They just describe different setups. We keep the slacks for two more pages, because they hand us the next result for free.

Key takeaway

A slack variable is a squared number that fills the gap to the fence. It turns $g_i \leq 0$ into $g_i + t_i^2 = 0$ with no loss.

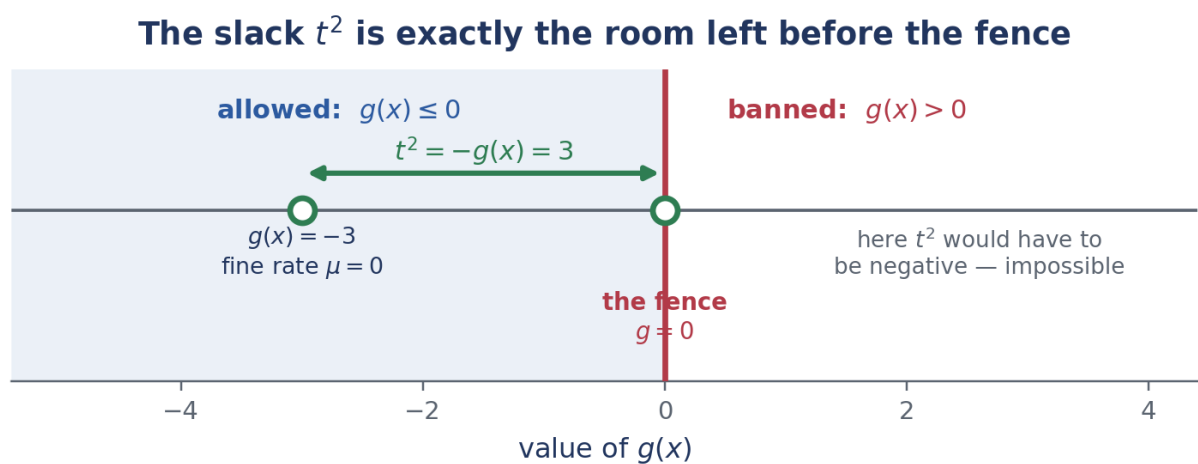


Figure 7: How to read it: the axis is the value of $g(x)$, and only the shaded left half is allowed. The green bar is t^2 , the room left before the fence. At the fence the bar has length zero.

7 Complementary slackness, derived in four lines

Differentiate the Lagrangian with respect to the room-left variable. Out drops the rule that a fine and spare room can never both be present.

This is the result most books state and never prove. With the slack variable in hand, the proof is four lines.

Differentiate (6) with respect to t_i . The only term containing t_i is $\mu_i t_i^2$:

$$\frac{\partial \mathcal{L}}{\partial t_i} = 2\mu_i t_i = 0. \quad (7)$$

Multiply both sides by t_i and halve:

$$\mu_i t_i^2 = 0. \quad (8)$$

From (5) we know $t_i^2 = -g_i(\mathbf{x})$. Substitute:

$$\mu_i (-g_i(\mathbf{x})) = 0 \quad \implies \quad \boxed{\mu_i g_i(\mathbf{x}^*) = 0} \quad (9)$$

That boxed line is **complementary slackness**. The name now makes sense: it is the statement about slack, and it was derived *from* the slack variable.

7.1 What it actually says

A product of two numbers is zero only when at least one of them is zero. So for each fence i , exactly one of these two worlds holds.

World 1: you have room to spare. Here $g_i(\mathbf{x}^*) < 0$. You are strictly inside the fence. The guard has no grounds to charge you, so $\mu_i = 0$. The fence is not doing any work. We say the fence is **inactive**, or *slack*.

World 2: you are touching the fence. Here $g_i(\mathbf{x}^*) = 0$. You are pressed against the edge. Now the fine rate is allowed to be positive, and usually is: $\mu_i > 0$. The fence is holding you back. We say the fence is **active**, or *tight*.

✗ Watch out

Two traps here, and both are easy to fall into.

Trap 1. “The fence is active” and “we are on the boundary” are the *same* world, not two different ones. Active means $g_i = 0$, which means on the boundary. The real split is: *fine is zero* versus *fence is touched*.

Trap 2. On the boundary the fine rate is *not* forced to zero. With $g_i = 0$ the product $\mu_i g_i$ is zero for *any* μ_i . In Section 11 you will meet a problem whose answer sits on the fence with $\mu = 4$. A positive fine on an active fence is the normal case, not an error.

✱ Wait — really?

The fine rate is not a bookkeeping device. It is a price. It measures how much the best score would improve if the fence were relaxed a little. Relax it by a small amount ε and the best score falls by about $\mu\varepsilon$. A fence with $\mu = 0$ is a fence you would pay nothing to remove, because it was never in your way.

$\mu g(x^*) = 0$ forces one of these two worlds

Room to spare

$$g(x^*) < 0$$

you are strictly inside the fence

the guard cannot charge you

so the fine rate is $\mu = 0$

this rule changes nothing

Touching the fence

$$g(x^*) = 0$$

you are pressed against the edge

the fine rate may be $\mu > 0$

this rule is holding you back

we call the rule ACTIVE

Figure 8: How to read it: the product $\mu_i g_i = 0$ leaves only two possible worlds per fence. Either the fine is zero, or the fence is touched. Never a positive fine with room to spare.

✓ Key takeaway

Complementary slackness: $\mu_i g_i(x^*) = 0$. Either the fine is zero, or the fence is touched. Never both loose at once.

8 The five KKT conditions

Differentiate the Lagrangian once with respect to every unknown, add the sign rule on the fines, and you have the full test for a constrained optimum.

The conditions are named after Karush, Kuhn and Tucker, so **KKT**. They are not a new idea. They are the old idea — set every derivative to zero — run on (6).

Work through the unknowns one at a time.

8.1 Condition 1: stationarity (from $\partial\mathcal{L}/\partial\mathbf{x}$)

Differentiate with respect to the n real variables. The slack terms t_i^2 do not contain $\mathbf{x}$, so they drop out:

$$\nabla f(\mathbf{x}^*) + \lambda \nabla e(\mathbf{x}^*) + \sum_{i=1}^m \mu_i \nabla g_i(\mathbf{x}^*) = \mathbf{0}. \quad (10)$$

This is n equations, one per variable. It is the arrows-on-one-line picture from (3), now written for many rules at once.

8.2 Condition 2: primal feasibility, path (from $\partial\mathcal{L}/\partial\lambda$)

Only the term $\lambda e(\mathbf{x})$ contains λ , and it is linear in it. So the derivative is just $e(\mathbf{x})$:

$$e(\mathbf{x}^*) = 0. \quad (11)$$

This is the path constraint, handed back to us. That is a good sign: the machine did not lose our rules.

8.3 Condition 3: primal feasibility, fences (from $\partial\mathcal{L}/\partial\mu_i$)

Only $\mu_i[g_i + t_i^2]$ contains μ_i , so

$$g_i(\mathbf{x}^*) + t_i^2 = 0 \quad \implies \quad g_i(\mathbf{x}^*) = -t_i^2 \leq 0.$$

So

$$g_i(\mathbf{x}^*) \leq 0, \quad i = 1, \dots, m. \quad (12)$$

Again, our own fences returned to us.

8.4 Condition 4: complementary slackness (from $\partial\mathcal{L}/\partial t_i$)

This is the result we proved above:

$$\mu_i g_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, m. \quad (13)$$

8.5 Condition 5: dual feasibility (the sign rule)

$$\mu_i \geq 0, \quad i = 1, \dots, m. \quad (14)$$

No derivative gives you this one. It came from Rule B: a fine must raise the bill, never lower it. Note that λ is *not* in this list. It stays free.

X Watch out

You will see KKT quoted as “four conditions” in some notes and “five” in others. The four-item version folds $\mu_i \geq 0$ into the setup instead of listing it. Nothing changes. Count five when you write an exam answer, so no marker can say you missed a check.

8.6 How many cases will you test?

Condition (13) is one product per fence. Each product offers two branches: the fine is zero, or the fence is touched. The branches multiply:

$$\text{number of cases} = 2^m.$$

So $m = 1$ gives 2 cases, $m = 2$ gives 4, $m = 3$ gives 8, and $m = 4$ gives 16.

✗ Watch out

It is easy to guess m^2 instead of 2^m . They agree at $m = 2$ (both give 4) and at $m = 4$ (both give 16), which hides the mistake. They disagree at $m = 3$: the true count is $2^3 = 8$, not 9. Always double the count for each new fence.

Three things about that count are worth knowing before you rely on it.

Equality constraints do not branch. The path is always tight, and λ is never forced to zero. So $e(\mathbf{x}) = 0$ adds no cases. Only the fences do.

The branches overlap. A point can have *both* $\mu_i = 0$ and $g_i = 0$ at once. That is the **degenerate** case, and it will show up in two different branches. So the 2^m cases cover everything, but they are not a clean split. Finding the same point twice is normal.

It counts cases, not answers. A branch may give no solution, exactly one, or a whole family. In Section 9 one branch dies on arithmetic before any fine rate is computed. So 2^m is the number of branches you must *check*, not the number of candidates you will collect.

* Wait — really?

2^m grows fast. Ten fences give 1024 branches, and a support vector machine has one fence per training point. That is why nobody solves a real SVM by listing cases. You solve the dual instead, which is Section 11. The case list is an exam tool, not an algorithm.

8.7 When is the KKT test trustworthy?

Two honest limits belong with the five conditions. Both are usually left out. First, some vocabulary. *Necessary* means every true answer must pass the test. *Sufficient* means anything that passes the test is a true answer. KKT is not automatically either one.

KKT is necessary only if the constraints are well behaved at the answer. The technical name is a **constraint qualification**. The common one is: the gradients of the active constraints at $\mathbf{x}^*$ must be linearly independent. Two active fences with parallel gradients break that. Then the multipliers may not exist at all, and the test can miss the true answer. Every problem in this reader passes, and so will your exam problems.

KKT is sufficient only if the problem is convex. A point that passes all five checks is a **KKT point**. For a convex problem — convex f , convex g_i , linear e — a KKT point is the global minimum, full stop. For a non-convex problem it may be a maximum or a saddle, so you must check it a second way.

✗ Watch out

Do not write “all five conditions hold, so this is the minimum” unless the problem is convex. Say which part is convex, or verify the point another way. Section 10 shows both routes on a non-convex constraint.

✓ Key takeaway

Five conditions: stationarity, path feasibility, fence feasibility, complementary slackness, and $\mu_i \geq 0$. Four fall out of derivatives; the fifth is the sign rule. They pin down the answer when the problem is convex.

9 Worked example: one path, two fences

We run all five KKT conditions on a problem with three rules, test all four cases, and reject three of them on signs alone.

 **Worked example:** Minimise $f(x, y) = x^2 + 2y^2$ subject to $x + y = 1, x \geq 0, y \geq 0$

Step 1. Standard form. The score is $f = x^2 + 2y^2$. The path is $e = x + y - 1 = 0$. The fences are $g_1 = -x \leq 0$ and $g_2 = -y \leq 0$. So $m = 2$, and we expect $2^2 = 4$ cases.

Step 2. Write the Lagrangian. Using (4):

$$\mathcal{L}(x, y, \lambda, \mu_1, \mu_2) = x^2 + 2y^2 + \lambda(x + y - 1) - \mu_1 x - \mu_2 y.$$

The minus signs come from $g_1 = -x$ and $g_2 = -y$. There are five unknowns.

Step 3. Stationarity in x . $\frac{\partial \mathcal{L}}{\partial x} = 2x + \lambda - \mu_1 = 0$, so

$$\mu_1 = 2x + \lambda. \quad (\text{E1})$$

Step 4. Stationarity in y . $\frac{\partial \mathcal{L}}{\partial y} = 4y + \lambda - \mu_2 = 0$, so

$$\mu_2 = 4y + \lambda. \quad (\text{E2})$$

Step 5. Path feasibility. $x + y - 1 = 0$, so

$$y = 1 - x. \quad (\text{E3})$$

Step 6. Get everything into x and λ . Put (E3) into (E2):

$$\mu_2 = 4(1 - x) + \lambda = 4 - 4x + \lambda. \quad (\text{E4})$$

Now (E1) and (E4) both use only x and λ . That was the goal.

Step 7. Fence feasibility and signs. We still need $x \geq 0, y \geq 0, \mu_1 \geq 0$ and $\mu_2 \geq 0$. These are the tests that will kill the bad cases.

Step 8. Complementary slackness. From (13), $\mu_1(-x) = 0$ and $\mu_2(-y) = 0$, which simplify to

$$\mu_1 x = 0, \quad \mu_2 y = 0. \quad (\text{E5})$$

Each gives two branches, so the four cases are: $(\mu_1=0, \mu_2=0)$, $(\mu_1=0, y=0)$, $(x=0, \mu_2=0)$, $(x=0, y=0)$.

 **Worked example:** The four cases, tested one by one

Step 1. Case 1: $\mu_1 = 0$ and $\mu_2 = 0$. Put $\mu_1 = 0$ in (E1): $2x + \lambda = 0$, so $\lambda = -2x$. Put $\mu_2 = 0$ in (E4): $4 - 4x + \lambda = 0$, so $\lambda = 4x - 4$. Both describe the same λ , so set them equal:

$$-2x = 4x - 4 \implies 6x = 4 \implies x = \frac{2}{3}.$$

Then $y = 1 - \frac{2}{3} = \frac{1}{3}$ and $\lambda = -2 \cdot \frac{2}{3} = -\frac{4}{3}$.

Step 2. Check Case 1 against every condition. $x = \frac{2}{3} \geq 0$ and $y = \frac{1}{3} \geq 0$, so both fences hold. $\mu_1 = \mu_2 = 0 \geq 0$, so dual feasibility holds. And $\frac{2}{3} + \frac{1}{3} = 1$, so the path holds. Case 1 survives. The score is

$$f = \left(\frac{2}{3}\right)^2 + 2\left(\frac{1}{3}\right)^2 = \frac{4}{9} + \frac{2}{9} = \frac{6}{9} = \frac{2}{3}.$$

Step 3. Case 2: $\mu_1 = 0$ and $y = 0$. From (E3), $y = 0$ forces $x = 1$. Put $x = 1$ into (E1) with $\mu_1 = 0$: $2(1) + \lambda = 0$, so $\lambda = -2$. Now read μ_2 off (E4): $\mu_2 = 4 - 4(1) + (-2) = -2$. But $\mu_2 \geq 0$ is required. Case 2 is rejected.

Step 4. Case 3: $x = 0$ and $\mu_2 = 0$. From (E3), $x = 0$ forces $y = 1$. Put $x = 0$ into (E4) with $\mu_2 = 0$: $4 - 0 + \lambda = 0$, so $\lambda = -4$. Now read μ_1 off (E1): $\mu_1 = 2(0) + (-4) = -4$. Again negative. Case 3 is rejected.

Step 5. Case 4: $x = 0$ and $y = 0$. Then $x + y = 0$. But the path needs $x + y = 1$, and $0 \neq 1$. Case 4 is rejected before any fine rate is computed.

Step 6. Answer. Only Case 1 survives:

$$\begin{aligned} x^* &= \frac{2}{3}, & y^* &= \frac{1}{3}, & \lambda^* &= -\frac{4}{3}, \\ \mu_1^* &= \mu_2^* = 0, & f^* &= \frac{2}{3}. \end{aligned}$$

Two things are worth reading off that answer.

First, $\lambda^* = -\frac{4}{3}$ is negative, and that is fine. Rule A allows it. If you had forced $\lambda \geq 0$ you would have found nothing.

Second, this problem is convex: $x^2 + 2y^2$ is a bowl, and all three rules are linear. So the one surviving KKT point is the global minimum. No second check is needed here.

Third, both fine rates came out zero. That says the answer sits strictly inside both fences: $x^* > 0$ and $y^* > 0$. Neither fence was doing any work. Once you see that, you could have solved the problem with the path alone. But you only *know* it after the check, so run the cases anyway.

⚠ Watch out

A common shortcut is to test only the case where all fines are zero, find an answer, and stop. That works here, but it will fail you when a fence really bites. Always test all 2^m cases, or say in writing why a case is impossible.

Smallest oval of $x^2 + 2y^2$ that still touches the line

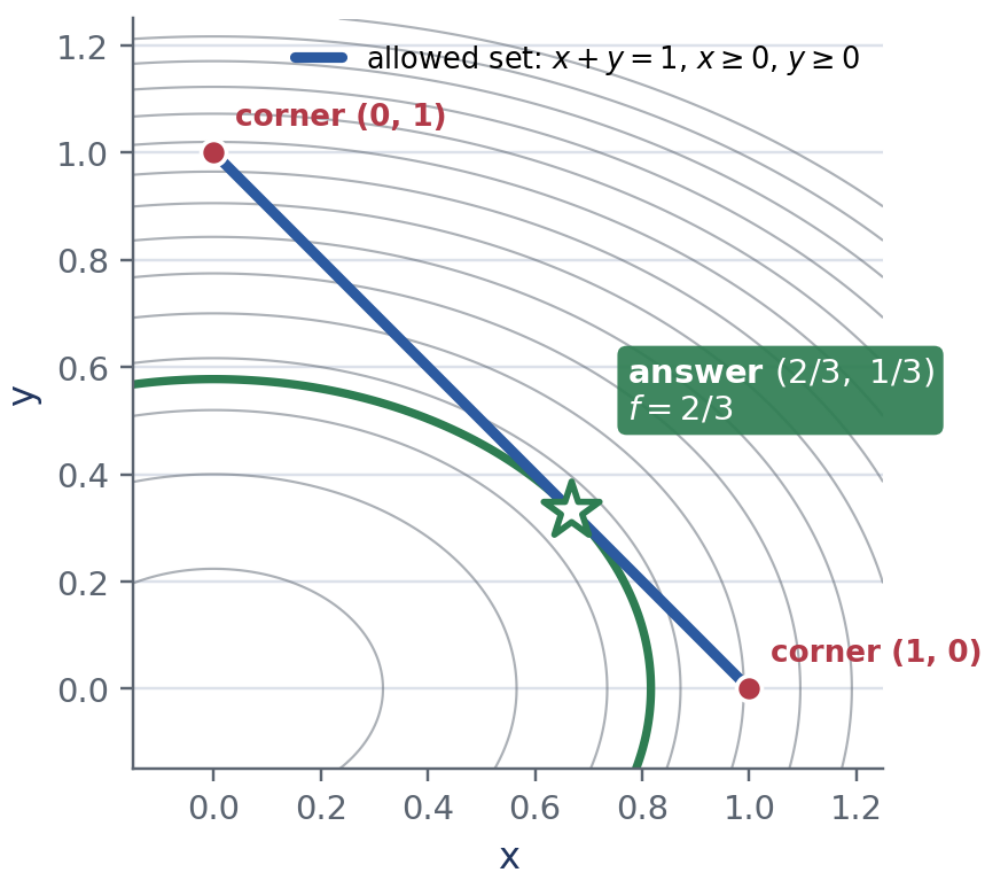


Figure 9: How to read it: grey rings are level curves of $x^2 + 2y^2$; the thick blue segment is the allowed set. Shrink the rings until the smallest one still touches the segment. It touches at the star, $(2/3, 1/3)$, where the value is $2/3$.

10 Two fast problems with a path and no fence

With only an equality constraint, KKT collapses to one line: set ∇f parallel to ∇e , then use the constraint to pin down the numbers.

Most exam questions have an equality constraint and nothing else. Then conditions 3, 4 and 5 are empty, and all you need is

$$\nabla f(\mathbf{x}^*) + \lambda \nabla e(\mathbf{x}^*) = \mathbf{0}, \quad e(\mathbf{x}^*) = 0. \quad (15)$$

Because λ is free, you may write $\nabla f = \lambda \nabla e$ instead. The value of λ flips sign and the answer does not move. The first example shows that happening.

 **Worked example: Maximise $f(x, y) = xy$ subject to $x + y = 6$**

Step 1. Set up. The path is $e = x + y - 6 = 0$.

Step 2. Two gradients. $\nabla f = (y, x)$ and $\nabla e = (1, 1)$.

Step 3. Make them parallel. Using $\nabla f = \lambda \nabla e$:

$$(y, x) = \lambda (1, 1) \implies y = \lambda \text{ and } x = \lambda.$$

Step 4. Use the path. Put $x = y = \lambda$ into $x + y = 6$: $2\lambda = 6$, so $\lambda = 3$, and $x = y = 3$.

Step 5. Read the value. $f = 3 \times 3 = 9$. That is the largest product of two numbers adding to 6.

Step 6. Now flip the sign. Using $\nabla f = -\lambda \nabla e$ instead gives $y = -\lambda$ and $x = -\lambda$, so $-2\lambda = 6$ and $\lambda = -3$. Then $x = -(-3) = 3$ and $y = 3$ again, with $f = 9$. Same point, same value, opposite λ . Rule A in action.

 **Worked example: Minimise $f(x, y) = x^2 + y^2$ subject to $xy = 1$**

Step 1. Set up. The path is $e = xy - 1 = 0$.

Step 2. Two gradients. $\nabla f = (2x, 2y)$ and $\nabla e = (y, x)$.

Step 3. Make them parallel. From $\nabla f + \lambda \nabla e = 0$:

$$2x + \lambda y = 0 \quad \text{and} \quad 2y + \lambda x = 0.$$

Step 4. Solve for λ . The first line gives $x = -\frac{\lambda y}{2}$. Put that into the second:

$$2y + \lambda \left(-\frac{\lambda y}{2} \right) = 0$$

$$2y - \frac{\lambda^2 y}{2} = 0$$

$$y \left(2 - \frac{\lambda^2}{2} \right) = 0.$$

Step 5. Rule out $y = 0$. If $y = 0$ then $xy = 0$, and the path needs $xy = 1$. So $y \neq 0$, and the bracket must vanish: $\lambda^2 = 4$, giving $\lambda = \pm 2$.

Step 6. Try $\lambda = -2$. Then $x = -\frac{(-2)y}{2} = y$. The path gives $xy = x^2 = 1$, so $x = \pm 1$. Two points work: $(1, 1)$ and $(-1, -1)$. Both give $f = 1^2 + 1^2 = 2$.

Step 7. Try $\lambda = +2$. Then $x = -y$, and the path becomes $xy = -y^2 = 1$. A square is never negative, so $-y^2$ is never $+1$. This branch has no solution.

Step 8. Check it, because this problem is not convex. The set $xy = 1$ is a hyperbola, not a line, so KKT alone does not prove we found a minimum. Verify directly. On the curve, $y = 1/x$, so

$$f = x^2 + \frac{1}{x^2}.$$

Write $u = x^2 > 0$. Then $f = u + 1/u$, and

$$u + \frac{1}{u} - 2 = \frac{u^2 - 2u + 1}{u} = \frac{(u - 1)^2}{u} \geq 0.$$

So $f \geq 2$ everywhere on the curve, with equality only at $u = 1$, which is $x = \pm 1$. The KKT point really is the minimum.

Step 9. Answer. The minimum is $f^* = 2$, reached at $(1, 1)$ and at $(-1, -1)$.

✓ **Key takeaway**

With only a path, the whole method is: set ∇f parallel to ∇e , solve for λ , then use the path to pin down the point.

11 Primal and dual: the same game, played in a different order

You and the guard both move. If you move first you are solving the primal. If the guard moves first you are solving the dual. The order changes the value, and never in your favour.

Look at the Lagrangian once more, for a single fence:

$$\mathcal{L}(x, \mu) = f(x) + \mu g(x), \quad \mu \geq 0.$$

Two players share this one number. You pick x and want the bill low. The guard picks μ and wants the bill high. That is a two-player, zero-sum game, the kind John Nash studied.

Now ask: *who moves first?*

11.1 The primal: you move first

If you pick x first, the guard sees your choice and answers with the worst μ for you. So the value of picking x is $\max_{\mu \geq 0} \mathcal{L}(x, \mu)$, and your best play is

$$p^* = \min_x \max_{\mu \geq 0} \mathcal{L}(x, \mu). \quad (16)$$

This is the **primal problem**. It is a min-max.

Here is why (16) is the original problem in disguise. Suppose you pick an x that breaks the fence, so $g(x) > 0$. The guard then raises μ without limit, and the inner maximum is $+\infty$. Any illegal x scores infinity, so you will never choose one. Suppose instead you obey the fence, so $g(x) \leq 0$. The best the guard can do is $\mu = 0$, and the value is just $f(x)$. So the min-max quietly rules out every illegal point and scores every legal one by its true score.

11.2 The dual: the guard moves first

Swap the order. The guard announces a fine rate μ , then you pick your best x knowing that price:

$$d^* = \max_{\mu \geq 0} \min_x \mathcal{L}(x, \mu). \quad (17)$$

This is the **dual problem**, a max-min. The inner part,

$$d(\mu) = \min_x \mathcal{L}(x, \mu),$$

is called the **dual function**. It has one input per fine rate, and no x at all.

★ Everyday picture

Announcing the price first is a worse deal for the guard. You get to see the number and plan around it. Moving second is an advantage, and the guard loses it. That is the whole reason d^* can only fall short of p^* .

11.3 Weak duality, strong duality, and the gap

Moving second is never worse, so the guard's value cannot beat yours:

$$d^* \leq p^*. \quad (18)$$

This always holds, for any problem at all. It is called **weak duality**. The shortfall $p^* - d^* \geq 0$ is the **duality gap**.

When the gap is zero, so $d^* = p^*$, we have **strong duality**. This happens for convex problems that have a strictly feasible point, which covers ridge, lasso and the support vector machine.

⚠ Watch out

Say it in the right direction. For a *minimisation* primal, the dual sits *below*: $d^* \leq p^*$. A positive gap is not “weak duality” as a special case of equality — weak duality is the inequality itself, and it is always true. Strong duality is the extra fact that the two meet.

11.4 Why anyone bothers

Three reasons, and the third is the one that matters in machine learning.

One: fewer unknowns. The primal has n variables. The dual has one per constraint. Swap a 10-variable primal for a 1-variable dual and the work collapses.

Two: the dual is always concave. Fix any x . Then $\mathcal{L}(x, \mu) = f(x) + \mu g(x)$ is a straight line in μ , with height $f(x)$ and slope $g(x)$. The dual $d(\mu)$ is the lowest of all those lines, one line per x . The lowest of a bundle of straight lines always bends downward, so d is concave. This holds even when f is not convex. So maximising d is a well-behaved problem with no local traps.

Three: the dual can change what the variables mean. In the support vector machine, the primal is over the weight vector, whose length is the number of features. The dual is over one μ_i per training point. That swap is what lets kernels enter, and it is why we spent this session here.

⚠ Watch out

“The dual is convex” is loose talk. The dual *function* is concave. The dual *problem* — maximise a concave function — is a convex optimization problem. Both statements are right, and they are about different objects.

 Worked example: Build a dual from scratch: minimise $(x - 3)^2$ subject to $x \leq 1$

Step 1. Standard form. The score is $f(x) = (x - 3)^2$. The fence is $g(x) = x - 1 \leq 0$.

Step 2. Solve the primal by eye. The bowl bottoms at $x = 3$, which is banned. So you slide left to the fence: $x^* = 1$ and

$$p^* = (1 - 3)^2 = 4.$$

Step 3. Write the Lagrangian.

$$\mathcal{L}(x, \mu) = (x - 3)^2 + \mu(x - 1), \quad \mu \geq 0.$$

Step 4. Inner minimum over x , with μ held fixed.

$$\frac{\partial \mathcal{L}}{\partial x} = 2(x - 3) + \mu = 0 \implies x = 3 - \frac{\mu}{2}.$$

The second derivative is $2 > 0$, so this is a minimum, not a maximum.

Step 5. Put that x back in to get the dual function.

$$\begin{aligned} d(\mu) &= \left(3 - \frac{\mu}{2} - 3\right)^2 + \mu\left(3 - \frac{\mu}{2} - 1\right) \\ &= \left(-\frac{\mu}{2}\right)^2 + \mu\left(2 - \frac{\mu}{2}\right) \\ &= \frac{\mu^2}{4} + 2\mu - \frac{\mu^2}{2} \\ &= 2\mu - \frac{\mu^2}{4}. \end{aligned}$$

Notice the shape: an upside-down parabola. Concave, exactly as promised.

Step 6. Maximise the dual over $\mu \geq 0$.

$$\frac{d}{d\mu} \left(2\mu - \frac{\mu^2}{4} \right) = 2 - \frac{\mu}{2} = 0 \implies \mu^* = 4.$$

That satisfies $\mu \geq 0$, so we keep it.

Step 7. Read the dual value.

$$d^* = 2(4) - \frac{4^2}{4} = 8 - 4 = 4.$$

Step 8. Compare. $p^* = 4$ and $d^* = 4$. The gap is zero, so this problem has strong duality.

Step 9. Recover the point. $x^* = 3 - \frac{4}{2} = 1$, which matches Step 2. And check complementary slackness: $\mu^* g(x^*) = 4 \times (1 - 1) = 0$. It holds.

Two different curves, one shared peak value: $4 = 4$

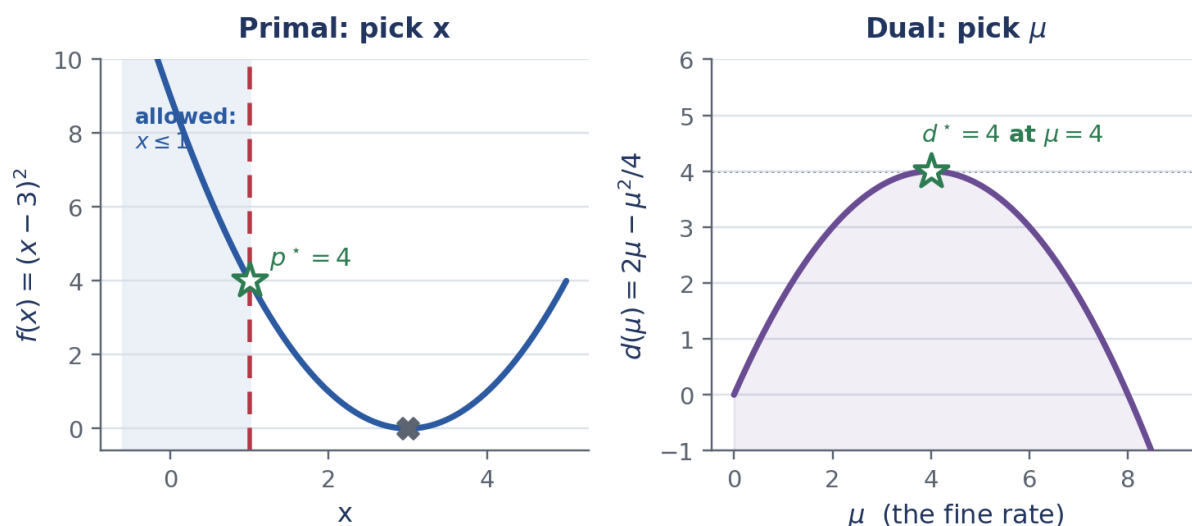


Figure 10: How to read it: two different curves over two different variables. On the left you choose x and the shaded strip is allowed. On the right you choose μ and the curve is concave. Their best values agree at 4.

* Wait — really?

That last example kills a tempting rule of thumb. The answer sits *on* the fence, and the fine rate is $\mu^* = 4$, not zero. On an active fence the fine rate is normally positive. It is only *inside* the fence that complementary slackness forces the fine to zero.

✓ Key takeaway

Primal is min then max. Dual is max then min. The dual never overshoots: $d^* \leq p^*$. When they meet, solve whichever one is easier.

12 The full problem: primal by KKT, then the dual, end to end

One problem, one path, one fence, both constraints biting. We solve it twice and watch the two answers land on the same number.

Here is the problem to keep. It has an equality constraint and an inequality constraint, and the inequality genuinely bites.

$$\begin{array}{ll} \text{Minimise} & f(x, y) = x^2 + y^2 \\ \text{subject to} & x + y = 4 \quad \text{and} \quad x \geq 3. \end{array}$$

In the park story: you want the point closest to the origin, you must stay on the marked path $x + y = 4$, and a fence blocks everything with $x < 3$.

Worked example: Part A — the primal, using all five KKT conditions

Step 1. Standard form. Path: $e = x + y - 4 = 0$. Fence: $x \geq 3$ becomes $g = 3 - x \leq 0$. One fence, so $m = 1$ and there are $2^1 = 2$ cases.

Step 2. Lagrangian.

$$\mathcal{L}(x, y, \lambda, \mu) = x^2 + y^2 + \lambda(x + y - 4) + \mu(3 - x), \quad \mu \geq 0.$$

Step 3. Stationarity. Differentiate in x and in y :

$$\frac{\partial \mathcal{L}}{\partial x} = 2x + \lambda - \mu = 0, \tag{S1}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y + \lambda = 0 \implies \lambda = -2y. \tag{S2}$$

Step 4. Case 1: the fine is zero, $\mu = 0$. Then (S1) gives $2x + \lambda = 0$, so $\lambda = -2x$. Compare with (S2), $\lambda = -2y$, to get $x = y$. The path $x + y = 4$ then gives $x = y = 2$. Check the fence: $g = 3 - 2 = 1 > 0$, so $x \geq 3$ is broken. Case 1 is rejected.

Step 5. Case 2: the fence is touched, $g = 0$. Then $3 - x = 0$, so $x = 3$. The path gives $y = 4 - 3 = 1$. From (S2), $\lambda = -2(1) = -2$. From (S1), $\mu = 2x + \lambda = 6 + (-2) = 4$.

Step 6. Check every condition for Case 2. Path: $3 + 1 = 4$. Holds. Fence: $g = 3 - 3 = 0 \leq 0$. Holds. Dual feasibility: $\mu = 4 \geq 0$. Holds. Complementary slackness: $\mu g = 4 \times 0 = 0$. Holds. Stationarity: both (S1) and (S2) were used to build the numbers.

Step 7. Primal answer.

$$x^* = 3, \quad y^* = 1, \quad \lambda^* = -2, \quad \mu^* = 4, \quad p^* = 3^2 + 1^2 = 10.$$

Now solve the same problem the other way. We build the dual function and maximise it.

Worked example: Part B — the dual, built and solved in full

Step 1. Start from the same Lagrangian.

$$\mathcal{L}(x, y, \lambda, \mu) = x^2 + y^2 + \lambda(x + y - 4) + \mu(3 - x).$$

Both rules bite: the answer sits at the corner (3, 1)

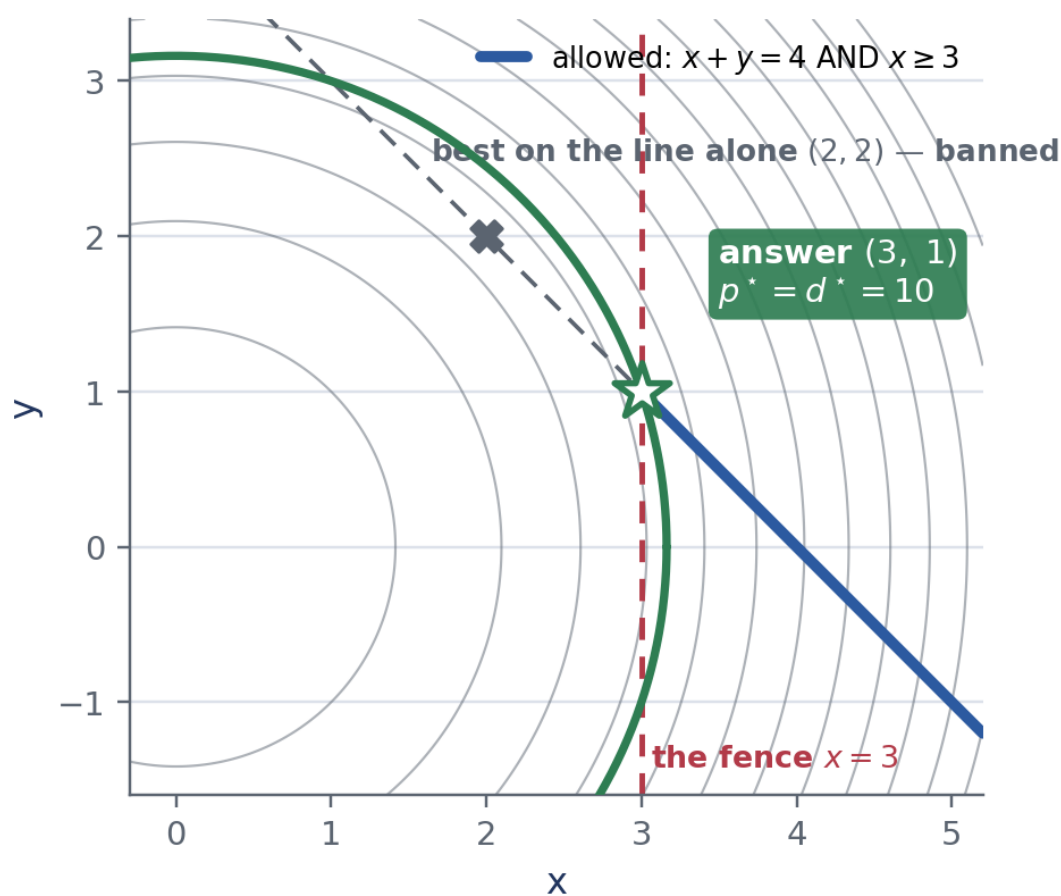


Figure 11: How to read it: grey rings are level curves of $x^2 + y^2$, so shrinking a ring means a better score. The dashed line is $x + y = 4$; only the blue part clears the fence $x = 3$. The green ring is the smallest one that still reaches the blue part, and it reaches it at the corner (3, 1).

Step 2. Minimise over x and y , holding λ and μ fixed. Setting both partial derivatives to zero:

$$\begin{aligned} 2x + \lambda - \mu &= 0 \implies x = \frac{\mu - \lambda}{2}, \\ 2y + \lambda &= 0 \implies y = -\frac{\lambda}{2}. \end{aligned}$$

The x, y part of $\mathcal{L}$ is $x^2 + y^2$ plus linear terms. Both squares have coefficient $+1$, so this really is the minimum.

Step 3. Substitute, piece by piece. Take the three groups separately.

$$\begin{aligned} x^2 + y^2 &= \frac{(\mu - \lambda)^2}{4} + \frac{\lambda^2}{4} = \frac{\mu^2}{4} - \frac{\lambda\mu}{2} + \frac{\lambda^2}{4} + \frac{\lambda^2}{4}, \\ x + y - 4 &= \frac{\mu - \lambda}{2} - \frac{\lambda}{2} - 4 = \frac{\mu}{2} - \lambda - 4, \\ 3 - x &= 3 - \frac{\mu - \lambda}{2} = 3 - \frac{\mu}{2} + \frac{\lambda}{2}. \end{aligned}$$

Step 4. Multiply out the two fine terms.

$$\begin{aligned} \lambda(x + y - 4) &= \frac{\lambda\mu}{2} - \lambda^2 - 4\lambda, \\ \mu(3 - x) &= 3\mu - \frac{\mu^2}{2} + \frac{\lambda\mu}{2}. \end{aligned}$$

Step 5. Add everything and collect like terms. Group the μ^2 , λ^2 and $\lambda\mu$ pieces:

$$\begin{aligned} \mu^2 : \quad & \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}, \\ \lambda^2 : \quad & \frac{1}{4} + \frac{1}{4} - 1 = -\frac{1}{2}, \\ \lambda\mu : \quad & -\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = +\frac{1}{2}. \end{aligned}$$

The leftover linear terms are -4λ and $+3\mu$. So the dual function is

$$d(\lambda, \mu) = -\frac{\mu^2}{4} - \frac{\lambda^2}{2} + \frac{\lambda\mu}{2} - 4\lambda + 3\mu. \quad (19)$$

Two variables gone. The dual is a problem in λ and μ only.

Step 6. Check that it is concave. The matrix of second derivatives of (19) is

$$H = \begin{pmatrix} -1 & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \quad \det H = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} > 0, \quad \text{trace } H = -\frac{3}{2} < 0.$$

A positive determinant with a negative trace means both eigenvalues are negative. So H is negative definite and d is concave. There is one peak and no local traps.

Step 7. Maximise: set both partial derivatives to zero.

$$\frac{\partial d}{\partial \lambda} = -\lambda + \frac{\mu}{2} - 4 = 0 \implies \lambda = \frac{\mu}{2} - 4, \quad (D1)$$

$$\frac{\partial d}{\partial \mu} = -\frac{\mu}{2} + \frac{\lambda}{2} + 3 = 0 \implies \mu = \lambda + 6. \quad (D2)$$

Step 8. Solve the two together. Put (D2) into (D1):

$$\lambda = \frac{\lambda + 6}{2} - 4 = \frac{\lambda}{2} + 3 - 4 = \frac{\lambda}{2} - 1.$$

So $\frac{\lambda}{2} = -1$, giving $\lambda^* = -2$. Then (D2) gives $\mu^* = -2 + 6 = 4$.

Step 9. Check the sign rule. We need $\mu \geq 0$, and $\mu^* = 4 \geq 0$. The peak lies inside the allowed region, so we keep it. (Had it come out negative, the best allowed μ would sit at the edge, $\mu = 0$.)

Step 10. Read the dual value. Put $\lambda = -2$ and $\mu = 4$ into (19):

$$\begin{aligned} d^* &= -\frac{16}{4} - \frac{4}{2} + \frac{(-2)(4)}{2} - 4(-2) + 3(4) \\ &= -4 - 2 - 4 + 8 + 12 \\ &= 10. \end{aligned}$$

Step 11. Compare with the primal. $p^* = 10$ and $d^* = 10$. The duality gap is zero, so strong duality holds.

Step 12. Recover the primal point from the dual answer. Use the formulas from Step 2:

$$\begin{aligned} x^* &= \frac{\mu^* - \lambda^*}{2} = \frac{4 - (-2)}{2} = 3, \\ y^* &= -\frac{\lambda^*}{2} = -\frac{-2}{2} = 1. \end{aligned}$$

The same $(3, 1)$ as Part A. The two routes agree at every number: $\lambda^* = -2$, $\mu^* = 4$, $x^* = 3$, $y^* = 1$, and the value 10.

* Wait — really?

Read the fine rate as a price. Here $\mu^* = 4$. Move the fence from $x \geq 3$ to $x \geq 2.9$ and the best score drops by about $4 \times 0.1 = 0.4$. Check it: the new answer is $(2.9, 1.1)$, giving $8.41 + 1.21 = 9.62$, which is 0.38 below 10. The multiplier told you the gain before you solved anything again.

x Watch out

Do not forget Step 9. Maximising the dual by derivatives finds the peak of the whole surface. If that peak has $\mu < 0$, it is out of bounds, and the best allowed value sits on the edge $\mu = 0$. Always test the sign before reporting d^* .

13 Test yourself

Nine questions covering every idea in this reader. Answers are printed upside down, so try each one first.

1. Why does standard form ban a strict $<$?
The best point may sit exactly on the boundary. A strict rule removes it and leaves values that keep improving with no smallest one, so the problem has no answer.
2. Convert " $x + 2y \geq 5$ " into standard form.
Multiply by -1 : $-x - 2y \leq -5$, so $g = -x - 2y + 5 \leq 0$.
3. A problem has three fences. How many cases does complementary slackness create?
Eight, from 2^3 . Not nine. Each fence gives two branches and the branches multiply.
4. At the answer, are ∇f and ∇g equal?
No, parallel. They lie on one line, related by $\nabla f = -\mu \nabla g$. The lengths usually differ.
5. You find $\mu_2 = -3$ while testing a case. What do you do?
Reject that case. Dual feasibility needs $\mu_i \geq 0$. A negative fine would reward crossing the fence.
6. You find $\lambda = -4$. What do you do?
Nothing. The path multiplier is free in sign. Only fence multipliers must be non-negative.
7. A fence is active at the answer. Must its fine rate be zero?
No. Active means $g = 0$, so $\mu g = 0$ holds for any μ . The fine rate is usually positive there. It is forced to zero only when $g < 0$.
8. Where does the name "complementary slackness" come from?
From the slack variable t_i . Setting $\partial \mathcal{L} / \partial t_i = 2\mu_i t_i = 0$, then multiplying by t_i and using $t_i^2 = -g_i$, gives $\mu_i g_i = 0$.
9. Primal value 12, dual value 12. What is that called?
Strong duality, with a zero duality gap. In general only $d^* \leq p^*$ is guaranteed.
10. A point passes all five KKT conditions. Is it the minimum?
Only if the problem is convex. Otherwise it is just a KKT point, and it could be a maximum or a saddle. Check it a second way.
11. Why can a point turn up in two of the 2^m cases?
Because the branches overlap. A degenerate point has $\mu_i = 0$ and $g_i = 0$ at the same time, so it satisfies both branches for that fence.

14 The pocket card

Everything in this reader on one page. If you memorise one thing, memorise this.

Step	What you write
1. Standard form	$\min f(\mathbf{x})$ with $e(\mathbf{x}) = 0, g_i(\mathbf{x}) \leq 0$. Flip any $\geq$ by multiplying by -1 .
2. Lagrangian	$\mathcal{L} = f + \lambda e + \sum_i \mu_i g_i$, with λ free and $\mu_i \geq 0$.
3. Stationarity	$\nabla f + \lambda \nabla e + \sum_i \mu_i \nabla g_i = \mathbf{0}$ (n equations).
4. Path feasibility	$e(\mathbf{x}^*) = 0$.
5. Fence feasibility	$g_i(\mathbf{x}^*) \leq 0$ for every i .
6. Complementary slackness	$\mu_i g_i(\mathbf{x}^*) = 0$ for every i . Split into 2^m cases.
7. Dual feasibility	$\mu_i \geq 0$. Any case with a negative μ_i is thrown out.
8. Test the cases	Keep only cases passing steps 4, 5 and 7. Compare the surviving scores.
9. Dual (if wanted)	$d(\lambda, \mu) = \min_{\mathbf{x}} \mathcal{L}$; then maximise d over λ free, $\mu_i \geq 0$.
10. Duality check	$d^* \leq p^*$ always. Equal means strong duality.
11. Say why it wins	Convex problem $\Rightarrow$ the KKT point is the global minimum. Not convex $\Rightarrow$ verify a second way.

14.1 Friendly name $\rightarrow$ textbook name

In this reader	Textbook term	Symbol
the score	objective function	$f(\mathbf{x})$
the path	equality constraint	$e(\mathbf{x}) = 0$
a fence	inequality constraint	$g_i(\mathbf{x}) \leq 0$
the fine rate	Lagrange multiplier	λ, μ_i
room left	slack variable	t_i
touching the fence	active (tight) constraint	$g_i = 0$
room to spare	inactive (slack) constraint	$g_i < 0$
the total bill	the Lagrangian	$\mathcal{L}$
the guard's best price	the dual optimum	d^*

14.2 Symbol cheat sheet

Symbol	Meaning
$\mathbf{x}$	the list of n variables you are solving for
$f(\mathbf{x})$	the objective; the number to make small
$e(\mathbf{x})$	the equality constraint function; some books write h
$g_i(\mathbf{x})$	inequality constraint i , written so that $g_i \leq 0$
m	how many inequality constraints there are
λ	multiplier on the equality; any sign
μ_i	multiplier on inequality i ; must satisfy $\mu_i \geq 0$
t_i	slack variable for fence i , with $t_i^2 = -g_i$
$\mathcal{L}$	the Lagrangian
∇f	gradient of f ; the vector of partial derivatives
p^*	the primal optimal value
d^*	the dual optimal value
$p^* - d^*$	the duality gap; always ≥ 0
* superscript	marks an optimal value, as in x^*

✓ Key takeaway

Fences move the answer onto the boundary. The Lagrangian prices each fence. KKT is just “all derivatives zero” plus a sign rule. Complementary slackness comes from the slack variable, and it says a fine and spare room never appear together. The dual reorders the game and never overshoots.